Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**TokenSmith: A local-first AI tutor for industrial workers that grounds answers in equipment operation manuals**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), **AI (√)**, Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Knowledge Services			
	Medium Category	Small Category		
	Human Resource Competency Development Services	**Intelligent Learning Support / Management Technologies**		
Concept of Proposed Technology	The proposed technology is a **local-first AI tutoring system** that helps operators, technicians, and maintenance staff learn directly from their own equipment operation manuals, standard operating procedures (SOPs), and safety documents, rather than from generic web data. Instead of sending sensitive plant information to a cloud service, the system runs on shop-floor PCs, rugged tablets, or laptops inside the plant network, and builds a structured knowledge store from the company's manuals, troubleshooting guides, checklists, and training handouts. When a worker asks a question: "Why did the pressure alarm trigger and what should I check first?"			

	the system: 1. Retrieves only the relevant pages, steps, and diagrams from the approved manuals, 2. Generates a clear, step-by-step explanation, and 3. Attaches span-level citations that point back to the exact paragraphs or figures in the original documents, so supervisors and workers can verify every claim. Under the hood, we adapt the TokenSmith architecture: manuals and SOPs are preprocessed into a Procedure Store with stable identifiers for each step, warning, and troubleshooting tree, plus metadata such as equipment type, task category, and required qualifications. When questions come in, a budgeted planner coordinates retrieval, re-ranking, evidence assembly, answer generation, and claim-level verification under strict device constraints (CPU, memory, latency). Over time, the system builds a **lightweight model of each worker's mastery** and typical errors. For example, which procedures they repeatedly look up or where they misinterpret alarms, and uses that to adjust explanations and highlight typical pitfalls, again grounded in the same manuals and SOPs used on the plant floor. For Korean SMEs, this technology becomes an **industrial learning assistant** that: • standardizes how procedures are explained across shifts and sites, • supports new hires and multi-skill training without large training teams, and • keeps operational know-how inside the company firewall.
Rationale for Joint R&D Project	Industrial deployment of this technology requires both advanced information-systems research and deep localization into Korean manufacturing and service environments**.** • The research institute brings the core architecture and algorithms for local-first, evidence-grounded tutoring: cost-based planning of retrieval-and-generation pipelines, structured knowledge stores with stable span identifiers, and mastery-aware tutoring logic. • The Korean SME partner brings knowledge of local industry sectors (manufacturing, logistics, utilities, after-sales service), typical equipment and manuals, integration with existing CMMS/LMS systems, and access to real sites for pilot deployment and commercialization. A joint R&D project is needed because: 1. **Manuals and SOPs differ by sector and plant:** The structure, language, and level of detail in Korean operation manuals (e.g., for process equipment, packaging lines, HVAC, logistics systems) differ significantly from university textbooks. We must co-design document preprocessing, tagging, and retrieval for these industrial document types. 2. **On-premise constraints are strict:** Many factories and logistics centers prefer or require **on-premise, offline-capable** solutions with restricted internet connectivity. The joint team must engineer and test the system under real-world industrial IT policies and hardware constraints. 3. **Workforce training and safety are business-critical:** Misinterpretation of procedures can cause downtime or safety incidents. Korean SME partners can help design realistic evaluation scenarios (e.g., start-up/shutdown procedures, changeovers, line clearance checks) so we optimize for the metrics that actually matter on the shop floor. 4. **Commercialization requires a local champion:** To become a sustainable product, the technology needs packaging, integration, support, and a sales strategy for Korean SMEs and mid-sized manufacturers. This is squarely in the Korean partner's domain.

(max 10 pages)

	Through TIPA, this collaboration accelerates transfer of cutting-edge AI tutoring research into **practical industrial learning tools** that strengthen the training and safety capabilities of Korean SMEs.
Key Words	1. Local-first industrial AI tutor 2. Operation manual and SOP grounding 3. Evidence-linked explanations and citations 4. Worker skill and error-pattern modeling

(max 10 pages)

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL) [2)]	Start of the Project	Level (3)	End of the Project	Level (6)

R&D Objectives	The joint R&D project aims to develop and pilot a **TokenSmith-based, local-first tutoring platform tailored to industrial workers and operation manuals**, with the following specific objectives:
	
	1. **Build an industrial "Procedure Store" for manuals and SOPs**
	  ○ Preprocess equipment manuals, SOPs, safety procedures, troubleshooting trees, and checklists into a structured store with stable step-level identifiers, warnings, and cross-references for each machine and process.
	2. **Develop a budgeted tutoring engine for plant-floor devices**
	  ○ Design a planner and execution engine that runs retrieval–generation–verification plans under explicit device budgets on shop-floor PCs and tablets, while always aligning responses to approved documents and abstaining when evidence is insufficient.
	3. **Model worker skills and recurring errors**
	  ○ Maintain a compact model of each worker's familiarity with procedures, typical failure points, and high-risk misunderstandings, based on their queries, follow-ups, and simple feedback.
	4. **Localize for Korean industrial and service SMEs**
	  ○ Adapt document handling, UI, terminology, and examples to Korean language, sector-specific jargon, and local regulatory requirements (safety labels, lockout/tagout practices, etc.).
	5. **Pilot and validate in real Korean industrial environments**
	  ○ Conduct pilots in Korean SMEs (e.g., discrete manufacturing, process plants, logistics, equipment service providers) to measure learning, error reduction, downtime implications, and user acceptance, and to refine commercialization plans.			

R&D Target Specifications	Target specifications, to be finalized with Korean partners, focus on **grounding, safety, performance, and impact on training**:
	
	1. **Grounded answer and procedure quality**
	  • ≥ 85% of atomic claims in answers are traceably supported by explicit citations into operation manuals or SOPs in evaluation scenarios.
	  • ≤ 5% of answers contain a verified contradiction with the approved procedure for that task.
	2. **Abstention and policy compliance**
	  • ≥ 95% of out-of-scope or policy-sensitive queries (e.g., requests to ignore safety steps or bypass interlocks) are handled with abstention or safe guidance, not fabricated suggestions.
	3. **On-premise performance**
	  • Median response time ≤ 5 seconds for typical "how-to" or "what-went-wrong" questions on representative Korean shop-floor hardware (e.g., mid-range PC or rugged tablet with 8–16 GB RAM).
	  • Stable operation within plant IT resource limits without requiring external cloud connectivity during normal use.
	4. **Worker training outcomes and operational impact**
	  • Demonstrated improvement in task-specific assessments (e.g., start-up steps, changeover procedure) by at least 0.3 standard deviations compared to a control group using manuals alone.
	  • Measurable reduction in **procedure-related helpdesk calls or repeated errors** (target to be set per site, e.g., 20–30% reduction over pilot period).
	5. **Maintainability and content updates**			

(max 10 pages)

	• Ability to update manuals and SOPs (e.g., after engineering changes) and reindex them within 24 hours while preserving stable identifiers for most existing citations.

R&D Plan	The project is structured over about three years, moving from lab-scale adaptation of TokenSmith concepts to **pilot deployment in Korean industrial settings**.
	
	Year 1: Core Architecture and Industrial Procedure Store (TRL 3 to 4)
	
	• Define the overall architecture for an industrial learning assistant, reusing the TokenSmith concepts of a **Course Store** and **Learner Store**, but specialized into a **Procedure Store** and **Worker Store**.
	
	• Implement the Procedure Store for a small number of representative equipment lines (e.g., a packaging line, a process unit, a warehouse system):
	  ○ Parse manuals and SOPs into step-level spans, warnings, preconditions, and troubleshooting branches.
	  ○ Attach metadata: equipment model, line, task type (start-up, shutdown, changeover, inspection).
	• Build baseline retrieval and answer generation modules operating fully on local devices, with simple claim-level verification and citations back into the manuals.
	• Perform lab evaluations using synthetic and anonymized industrial scenarios, validating that the system respects device budgets and consistently links answers to approved content.
	
	Year 1 Milestones
	
	• Functional prototype that can answer “how-to” questions for one or two pilot machines using only their manuals and SOPs, with span-level citations.
	• Demonstrated end-to-end pipeline running under target latency and memory budgets on representative hardware in a test lab.
	
	Year 2: Worker Modeling, Korean Localization, and SME Integration (TRL 4 to 5)
	
	• Design a **Worker Store** for tracking which procedures each worker has consulted, common errors, and any supervisor feedback, in a privacy-respecting and auditable manner.
	• Incorporate this worker state into the tutoring engine so explanations, reminders, and caution notes adapt to the worker’s experience level and history.
	• Collaborate with the Korean SME to:
	  ○ Ingest Korean-language manuals, SOPs, and maintenance logs from at least two industrial clients.
	  ○ Localize the user interface, warning wording, and search phrases to match how Korean workers actually talk about equipment and alarms.
	  ○ Integrate the tutoring system with the SME’s platforms (e.g., LMS, CMMS, MES dashboards) and authentication schemes used in Korean plants.
	• Run **limited pilots at one or two Korean client sites**, focusing on a few high-value tasks (e.g., changeovers, periodic inspections, common alarm handling).
	
	Year 2 Milestones
	
	• Mastery-aware tutoring in at least one Korean industrial setting, with logs showing adaptation of explanations over time.

(max 10 pages)

	• Integrated prototype deployed through the SME's system for a small pilot group of workers, with initial feedback collected.
	
	Year 3: Multi-site Pilots, Safety/Impact Evaluation, and Commercial Readiness (TRL 5 to 6)
	
	• Expand pilots to multiple Korean industrial SMEs across different sectors (e.g., discrete manufacturing, logistics, utilities or facility management).
	• Conduct structured evaluations on:
	  ○ Worker learning and retention (short quizzes, scenario-based assessments).
	  ○ Frequency and seriousness of procedure-related errors before/after deployment.
	  ○ Worker and supervisor satisfaction, perceived usefulness, and trust.
	• Refine optimization, caching, and content update mechanisms based on pilot findings and feedback from Korean partners.
	
	• Work with the SME to define a product line and commercialization strategy (licensing, per-site deployment, support packages), including documentation and training material for customer IT and safety teams.
	
	Year 3 Milestones
	
	• Pilot deployments at multiple Korean customer sites, with evidence that target specifications on grounding, abstention, and latency can be met in real operations.
	• Commercialization-ready prototype with deployment guides, evaluation reports, and agreed-upon IP and support arrangements.

(max 10 pages)

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The research institute will focus on the underlying **information-systems and AI algorithms** and provide technical leadership: • Design of the core architecture (Procedure Store, Worker Store, planner, execution engine, caching and provenance mechanisms). • Development of cost-based planning for retrieval–generation–verification pipelines under strict device budgets, including safe reuse of intermediate artifacts. • Methods for structuring and indexing manuals, SOPs, and troubleshooting trees with stable span identifiers and cross-references. • Design of worker-state modeling approaches that can use interaction traces and simple feedback to improve future tutoring without violating safety or privacy constraints. • Design of evaluation protocols, metrics, and analysis for grounding quality, device-level performance, and training impact. • Technical guidance to the Korean SME's engineers through joint workshops, code reviews, and co-development sessions.
Expected R&R of Korean SME Partner	The Korean SME partner will focus on industrial adaptation, integration, and commercialization: • Identify target sectors and representative pilot customers (e.g., factories, logistics centers, service providers), and provide access to real manuals, SOPs, and training workflows. • Lead productization: integration of the research institute's modules into the SME's software stack; development of plant-ready interfaces (web/mobile) and dashboards for supervisors and training managers. • Ensure that Korean industrial terminology, safety labeling, and company policies are reflected correctly in the system's behavior and interfaces. • Manage installation, operation, and support during pilots, including coordination with customer IT and safety/quality teams. • Collect and structure feedback and operational data from pilots, and channel them back into joint improvement cycles. • Develop business models and marketing strategies for Korean SMEs, including pricing, support, and potential expansion to international customers.

(max 10 pages)

5. Expected Outcomes/Impact

Expected Outcomes and Impact	This joint project is expected to deliver:
	
	• **Technological Outcomes**
	  ○ A TokenSmith-inspired, local-first tutoring system adapted to industrial manuals and SOPs, running on plant-floor hardware with strong grounding and provenance guarantees.
	  ○ Reusable components (Procedure Store, Worker Store, planner, verification tools) that can be extended to additional machines, plants, and sectors.
	• **Industrial Training and Safety Impact**
	  ○ More consistent and traceable explanations of procedures across shifts and sites, helping new hires and multi-skilled workers come up to speed more quickly.
	  ○ Reduced dependence on "informal experts" for basic operational questions, freeing them to focus on complex problems.
	  ○ Potential reductions in procedure-related incidents, rework, and unplanned downtime.
	• **Commercialization Potential**
	  ○ A differentiated product for the Korean SME partner: an AI-based industrial learning assistant that emphasizes on-premise deployment, document-based grounding, and safety-aware behavior, aligned with Korean regulations and customer expectations.
	  ○ A path to expansion into other markets that face similar constraints (e.g., factories in other Asian countries with strict data policies).
	• **Broader Economic and Scientific Impact**
	  ○ Strengthened capabilities of Korean SMEs in AI-based knowledge services for industrial training, aligned with the Knowledge Services and AI categories in the national technology strategy.
	
	A concrete example of how local-first, evidence-grounded AI can be applied beyond education into industrial contexts, with joint publications and anonymized case studies.

6. References

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